

Khulna University of Engineering & Technology

Course No: CSE 4109

Course Title: Artificial Intelligence

Assignment No: 01

Assignment Topic: Modern Heuristic Search Techniques

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Year: Fourth

Semester: First

1. Selected Research Paper

The paper titled “*Dynamic Weighted Heuristic Trust Path Search Algorithm*” has been selected as it is directly related to my topic, that is it improves on classical A*.

Title: Dynamic Weighted Heuristic Trust Path Search Algorithm

Authors: Ru Kong and Xiangrong Tong

Published: IEEE Access, Volume 8, 2020

DOI: 10.1109/ACCESS.2020.3019797

2. Problem Statement

The paper addresses finding optimal trust paths between indirectly connected users in social networks. Existing approaches either consider all paths (too slow for large networks) or use strict limits (missing good solutions). Classical A* finds optimal paths but wastes time comparing nearly identical paths. Additionally, path length alone doesn't determine trust quality. The challenge is balancing accuracy with speed while considering both path length and trust consistency for real-time recommendation systems.

3. Proposed Methodology and Contribution

The authors of the paper developed the Dynamic Weighted Heuristic Trust Path Search (DWHS) algorithm with three key innovations that improve upon classical A*:

a. Six-Degree Space Constraint: The algorithm limits the search to paths with six hops or fewer based on six-degree separation theory. It uses BFS to find all the admissible paths. This reduces the search space dramatically while retaining practically useful paths.

b. Dynamic Weighting with FOCAL List: Instead of always selecting the single best node like A*, DWHS introduces a FOCAL list containing nodes that are “good enough” based on a dynamic weight factor:

$$f(n) \geq [w + (1-w) \times d(n)/N] \times f(n)_{\text{max}}.$$

The constraint relaxes for shallow nodes and tightens for deeper nodes. It avoids wasted comparisons between similar paths.

c. Secondary MSE Heuristic: From the FOCAL list, the algorithm selects the node with the smallest Mean Squared Error (MSE) of trust values along its path. This ensures stable, consistent trust paths and avoids routes with extreme value fluctuations that indicate unreliability.

The algorithm maintains OPEN, FOCAL, and CLOSED lists. So it dynamically filters candidates and expands the most promising nodes until reaching the target. Then it calculates the final trust value using the minimum function along the optimal path.

4. Results

The authors evaluated DWHS on two real-world datasets, Advogato (6,500 users with four trust levels) and FilmTrust (movie recommendations with trust ratings 1-10).

The key findings include:

- DWHS matched A*'s accuracy on Advogato ($w=0.9$) and significantly reduced running time. It achieved near-optimal solutions on FilmTrust ($w=0.87$) with much faster execution.
- DWHS consistently outperformed TidalTrust and MoleTrust across all evaluation metrics including Mean Absolute Error, Precision, Recall, and F-score.
- Parameter “w” enables tuning between solution quality and computational speed
- The MSE-based secondary heuristic successfully identified reliable trust paths. It produced more accurate predictions.

The authors concluded that DWHS effectively balances solution quality with computational efficiency. So it is practical for real-time trust prediction in large-scale recommendation systems.

5. Connection with Course Topics

Similarities to A* Algorithm:

DWHS follows the core structure of classical A*. It uses the same evaluation function:

$$f(n) = g(n) + h(n)$$

where,

$g(n)$ is the accumulated trust from the start to the current node,

$h(n)$ estimates the remaining trust to reach the target.

This combines actual progress with intelligent prediction, just like A* does for pathfinding.

- Both algorithms work with an OPEN list (nodes waiting to be explored).
- Both algorithms work with an CLOSED list (already processed nodes).
- They expand nodes based on their evaluation scores and guarantee finding a path if one exists.
- Both rely on domain knowledge. For DWHS the domain knowledge is trust values, to guide the search toward promising paths rather than blindly exploring everything.

Key Differences from Classical A*

The main differences are:

- **FOCAL List:** The biggest difference is the FOCAL list concept. Classical A* always picks the single best node from the OPEN list and expands it. DWHS instead identifies multiple "good enough" nodes based on a dynamic weight formula, puts them in a FOCAL list, and then uses MSE as a tiebreaker to choose the best one. This avoids wasting time comparing nearly identical paths.
- **Dynamic Weighting:** The dynamic weighting mechanism is another major departure. Standard A* treats all nodes equally regardless of depth. DWHS is more flexible with nodes close to the start (where many paths might work) and stricter with nodes deeper in the search (where choices matter more).
- **Limits Search:** DWHS limits the search to six hops based on the six-degree separation theory, while A* searches indefinitely until finding the optimal solution. This bounded approach trades theoretical perfection for practical speed, ensuring the algorithm finishes quickly even in massive networks.

Relation to Other Search Algorithms

Relation with BFS: DWHS uses BFS in the first phase to find all admissible paths within six hops. Like the BFS, it explores level by level, ensuring complete coverage at each depth.

However, unlike standard BFS that treats all edges equally, DWHS considers trust values as edge weights.

Relation with DFS: The algorithm avoids DFS because it would dive deep into one branch before checking alternatives. This could miss better paths at shallow levels, which contradicts the six-degree constraint that requires level-by-level exploration.

Relation with Hill Climbing: Compared to Hill Climbing, which makes locally optimal choices without backtracking, DWHS is more flexible. It maintains multiple candidate nodes in the FOCAL list and can reconsider options. It provides better global search while staying efficient. Hill Climbing might get stuck on a suboptimal path, but DWHS keeps exploring promising alternatives.

6. Improvements over Classical Methods

DWHS addresses several practical limitations of classical A* for real-world applications:

- **Time Efficiency:** Classical A* wastes time comparing paths with nearly identical quality (like 0.85 vs 0.84 vs 0.83). DWHS groups similar-quality nodes in the FOCAL list and uses MSE to choose quickly among them. So it shows dramatic speed improvements while maintaining near-identical accuracy.
- **Memory Efficiency:** The six-hop limit caps memory usage at a predictable level. Classical A* can grow exponentially as it searches deeper, but DWHS ensures bounded memory consumption, making it deployable in production systems.
- **Heuristic Quality:** The dual heuristic approach is smarter. While the primary heuristic finds high-trust paths, the MSE criterion ensures consistency, catching paths with one dangerously low value that A* would miss.
- **Real-World Robustness:** Trust values are noisy and subjective in practice. The MSE criterion prefers stable, consistent paths over erratic ones.

7. Suggestions for Further Improvement

While DWHS is a strong algorithm, several enhancements could make it even better:

Adaptive Parameter Tuning: Currently, w must be set manually for each dataset through trial and error. An automatic tuning system using Machine learning could adjust w dynamically during search, lowering it when the FOCAL list has too few nodes, raising it when there are too many.

Smarter Heuristics: The current heuristic could incorporate more information like user similarity, community structure, or trust patterns over time. Graph neural networks could learn better heuristics by training on many networks.

Multiple Path Output: Instead of returning just one path, finding the top- k diverse paths would provide backup options if intermediate users leave the network.

Parallel Processing: The algorithm explores one node at a time, but modern computers have multiple cores. Since the FOCAL list contains several good candidates, these could be explored simultaneously in parallel threads.

Reference:

1. R. Kong and X. Tong, "Dynamic Weighted Heuristic Trust Path Search Algorithm," in IEEE Access, vol. 8, pp. 157382-157390, 2020, doi: 10.1109/ACCESS.2020.3019797.